Adversarial Robustness of LSTM-Based Trading Systems

Six experiments exploring targeted attacks against an LSTM trading model

OVERVIEW

What this presentation walks through

This summary presents six experiments exploring targeted attacks against an LSTM trading model, each building on the previous one by answering a new question:

1. When is an attack most likely to succeed depending on timing policy and trading strategy?
2. Does retraining the model make it more robust?
3. Do the results still hold once a proper validation period is introduced?
4. A shift in perspective: what if success is redefined as an actual loss in cumulative return, not just a flipped signal?
5. Finally, can machine learning automate the search for the optimal day to launch a targeted attack?

EXPERIMENT 1 — TIMING POLICIES

The first experiment: does WHEN you attack matter?

Five attack-timing strategies were compared, ranging from random days, through market indicators, to exploiting the model's own decision boundary.

Three trading strategies were tested: Moving Average, Rate of Change, Bollinger Bands.

100%

success for Moving Average, but only in a narrow initial setup.

0%

non-trivial successes for Bollinger Bands and Rate of Change.

Timing matters, but only if it's strategy-aware. Attacks timed near the model's own decision boundary succeeded; generic market signals like volatility or trend failed to locate vulnerable days.

These results didn't generalize → Rate of Change became attackable in some broader configurations.

EXPERIMENT 2 — COMPARING LSTM TRAINING REGIMES

Does HOW the model is trained change how easy it is to attack?

Four training regimes were compared across every strategy and timing policy: fixed and retrained periodically.



Takeaway: frequent retraining on recent data cut attack success ten-fold versus the baseline.

EXPERIMENT 3 — FIXING A METHODOLOGICAL GAP

Correction confirms earlier findings

In the first two experiments, the attack was tested on the same time window used to evaluate the model's trading profitability. To correct this, a validation gap was introduced.

Setup	Attack success — same-period test (Exp. 2)	Attack success — held-out test (Exp. 3)
LSTM 3y	27.8%	**13.9%**
LSTM 13y	27.3%	**18.1%**
LSTM cumulative	20.8%	**12.3%**
LSTM rolling	2.7%	**6.9%**

Takeaway: success dropped out-of-sample, but the core finding held → Moving Average stayed fragile, rolling retrain stayed most robust.

EXPERIMENT 4 — DOES A FLIPPED SIGNAL ACTUALLY COST MONEY?

Success isn't just about flipping a signal

Every experiment so far measured "success" as: did the attack change the BUY/SELL signal relative to what the model would have said without interference? A new metric checks whether a flipped signal actually produced a negative cumulative return , over 10, 20, or ~750 days.

10 days after the attack

Financial success mostly 0–40%, often near zero. A flipped signal rarely hurts short-term returns.

20 days after the attack

Broadly similar to 10 days, with a slight rise for some setups.

Full ~3-year horizon

Financial success climbs sharply: up to 78% for some setups.

Takeaway: seemingly harmless attacks can accumulate into significant losses over time, while short-term monitoring may fail to capture their impact. Conversely, successfully executing an attack within a short time window can also be challenging for the attacker.

Exploring automated attack timing

So far, the best timing policy (near_strategy_boundary) required inside knowledge of the trading model. Can a separate classifier, trained on market data, learn to spot good attack days on its own? A dataset of past attacks and their outcomes was built, and several classifiers were trained and compared.

141

market-derived features engineered per day (returns, volatility, RSI, and more).

0.38

best PR-AUC (Random Forest): a weak score.

0.52

F1-score on the "attack" class: the model over-predicts vulnerable days.

Takeaway: the two most influential features for the model were the attack's own stated objective (force_buy / force_sell), not any market condition. Little real signal was found.

EXPERIMENT 6 — GIVING THE MODEL MORE DATA TO LEARN FROM

More training diversity didn't close the gap

The classifier was retrained with more scenarios and stocks, then the model was then deployed as its own "timing policy" and compared head-to-head against the timing policies from Experiment 1 on a fresh, unseen test period.

29%

of ML-selected attacks flipped the signal, the 2nd-best timing policy tested.

14%

of those attacks caused a real loss over the following 10–20 days, the 2nd-worst policy for actual impact.

Central discovery: flipping a decision and damaging a portfolio are separate skills → good at one doesn't mean good at the other.

Conclusions

- **Strategy matters more than market conditions**

  The trading strategy strongly affects vulnerability, while market-based timing policies struggle to reliably identify the most vulnerable days.

- **How a model is kept updated matters as much as its architecture.**

  Frequent retraining on recent data was consistently the hardest setup to attack.

- **Flipping a decision and causing financial harm are two different problems.**

  Many signal-flipping attacks barely hurt returns short-term; real damage often appears later.

- **Predicting truly damaging attacks from public data remains unsolved.**

  Both a hand-crafted heuristic and an ML classifier flip signals well, but neither reliably predicts real harm.

VALUE FOR THE STATE OF KNOWLEDGE

What this adds to what we know and what's still open

What this work contributes

A systematic map of ATS vulnerability across strategy, retraining, and timing, showing that signal-flipping and financial impact must be measured separately.

What remains open

Unclear whether the absence of predictive signal in Experiments 5–6 reflects a genuine lack of information in public market data, or simply a limitation of the features and models tried so far.